

***AI503- Machine Learning
Lecture 3:K-Nearest Neighbors
(KNN) Algorithm in Machine
Learning***

Manar Alkhatib

Supervised Learning for Classification and Regression

By the end of this lecture, students will understand:

- What KNN is and how it works
- Difference between classification and regression using KNN
- Distance metrics used in KNN
- How to choose optimal K value
- Bias vs Variance tradeoff
- Advantages and limitations
- Real-world applications
- Python implementation
- Performance evaluation

What is KNN?

The **K-Nearest Neighbors (KNN)** algorithm is a **supervised machine learning algorithm** used for:

- Classification problems
- Regression problems

It works by:

1. Finding the nearest K data points
2. Using their labels to predict the new sample

KNN is called a **lazy learner** because it does not build a model during training.

K-Nearest Neighbors

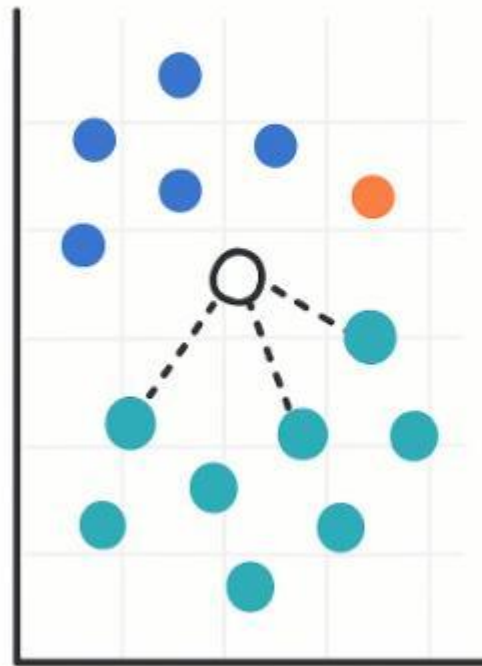
Supervised
Learning

Lazy
Algorithm

Based on
feature
similarity

Non
Parametric

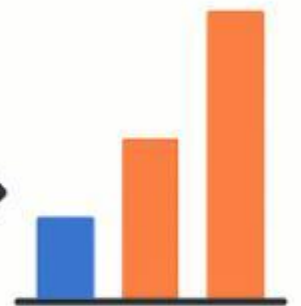
Query Point



Neighbors



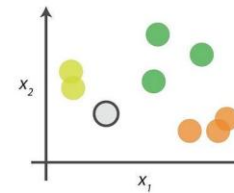
Class Vote



k-Nearest

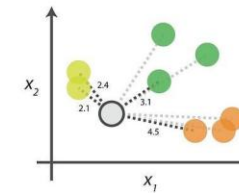
kNN Algorithm

0. Look at the data



Say you want to classify the grey point into a class. Here, there are three potential classes - lime green, green and orange.

1. Calculate distances



Start by calculating the distances between the grey point and all other points.

2. Find neighbours

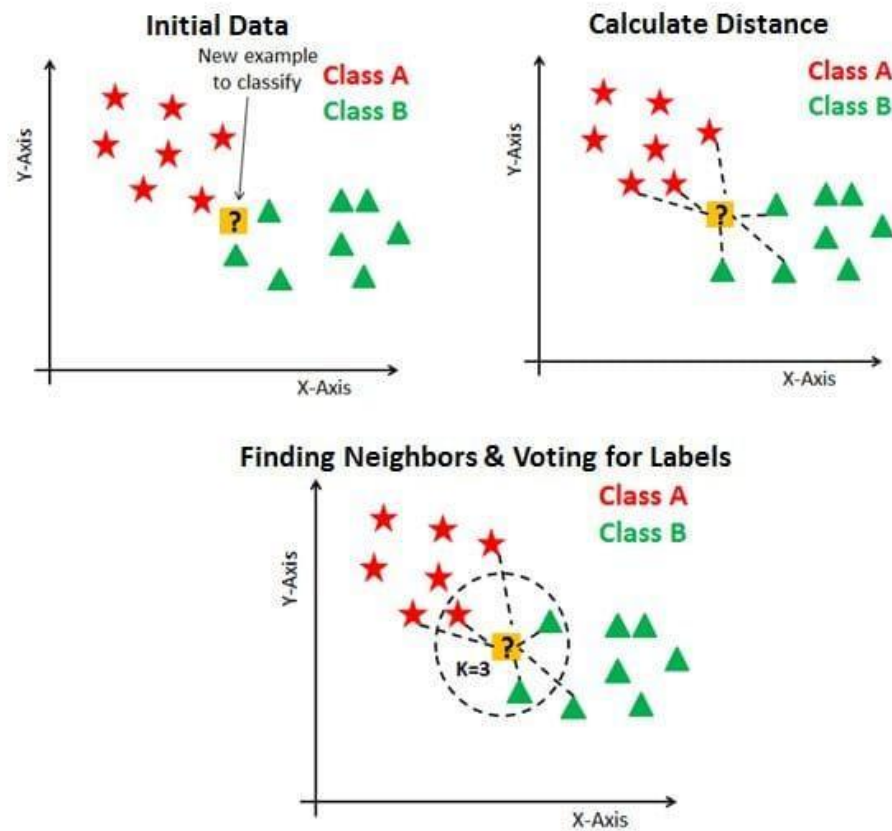
Point Distance		
	2.1	→ 1st NN
	2.4	→ 2nd NN
	3.1	→ 3rd NN
	4.5	→ 4th NN

Next, find the nearest neighbours by ranking points by increasing distance. The nearest neighbours (NNs) of the grey point are the ones closest in dataspace.

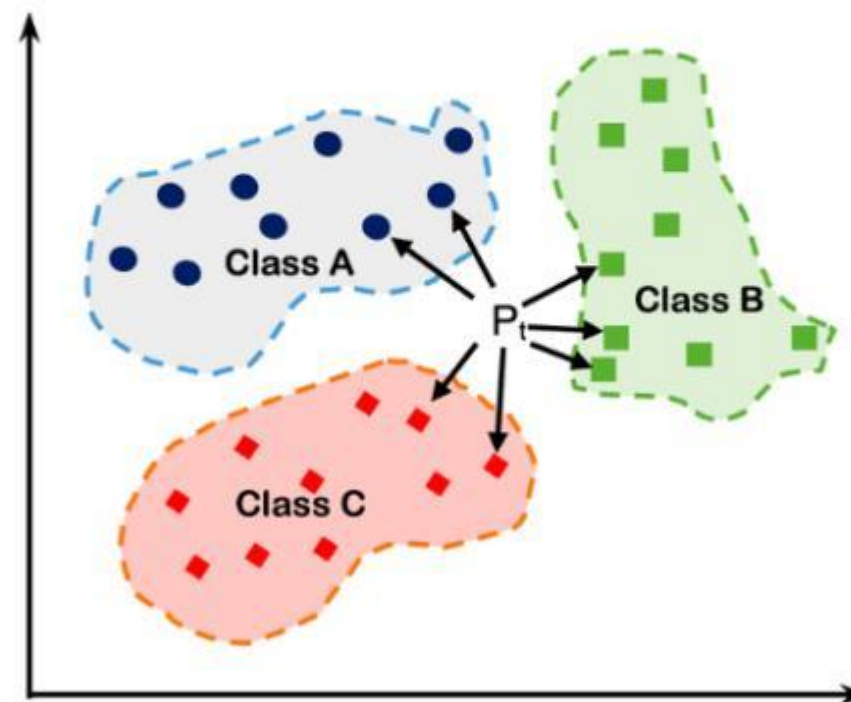
3. Vote on labels

Class	# of votes	
	2	→ Class wins the vote! Point is therefore predicted to be of class .
	1	
	1	

Vote on the predicted class labels based on the classes of the k nearest neighbours. Here, the labels were predicted based on the k=3 nearest neighbours.



K Nearest Neighbors



Why is it Called KNN?

- **K** = Number of neighbors considered
- **Nearest** = Based on distance
- **Neighbors** = Closest training samples

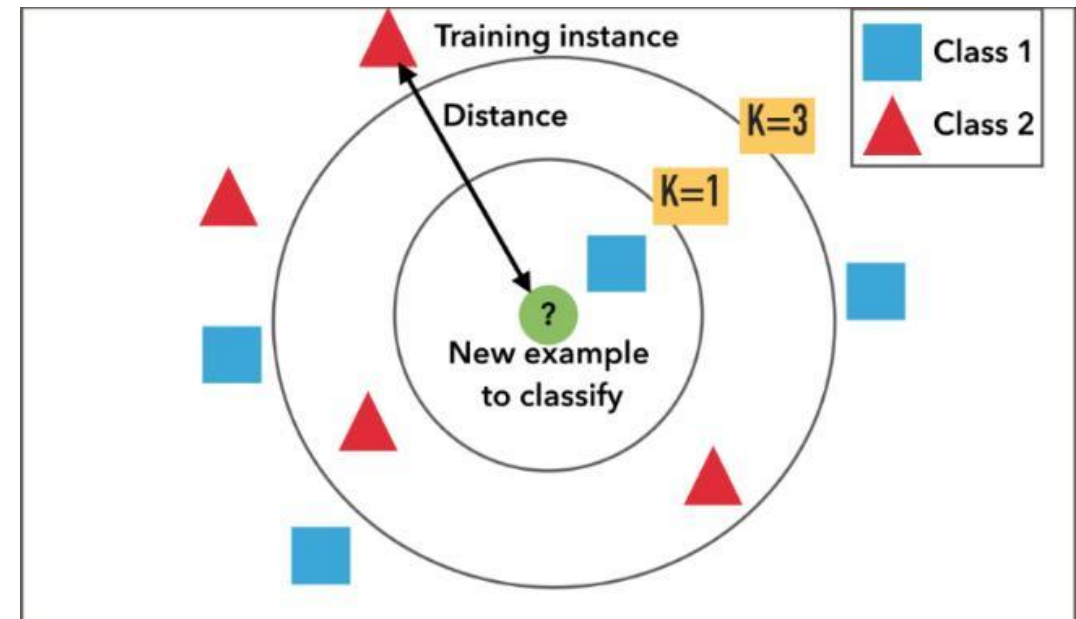
Example:

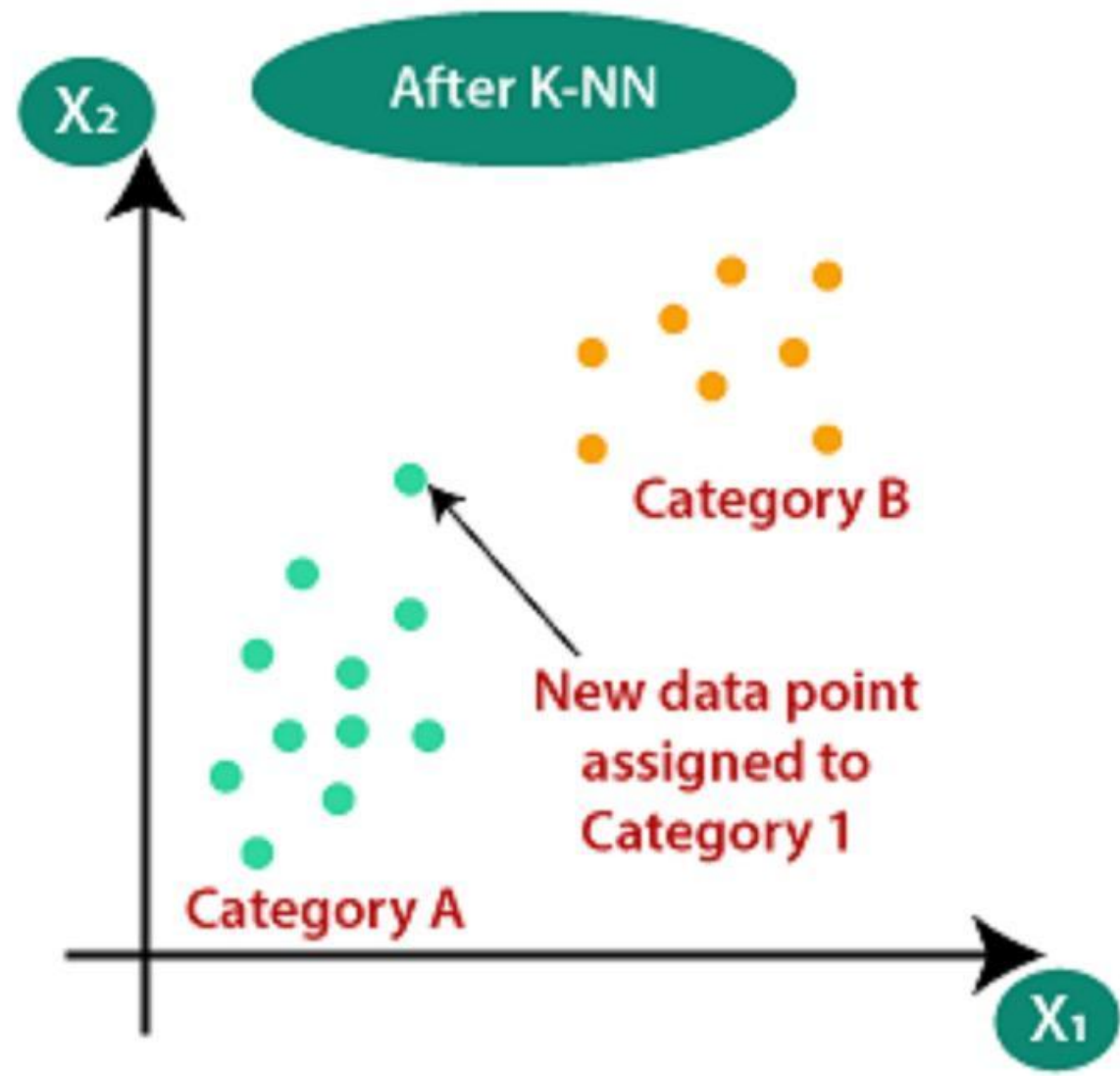
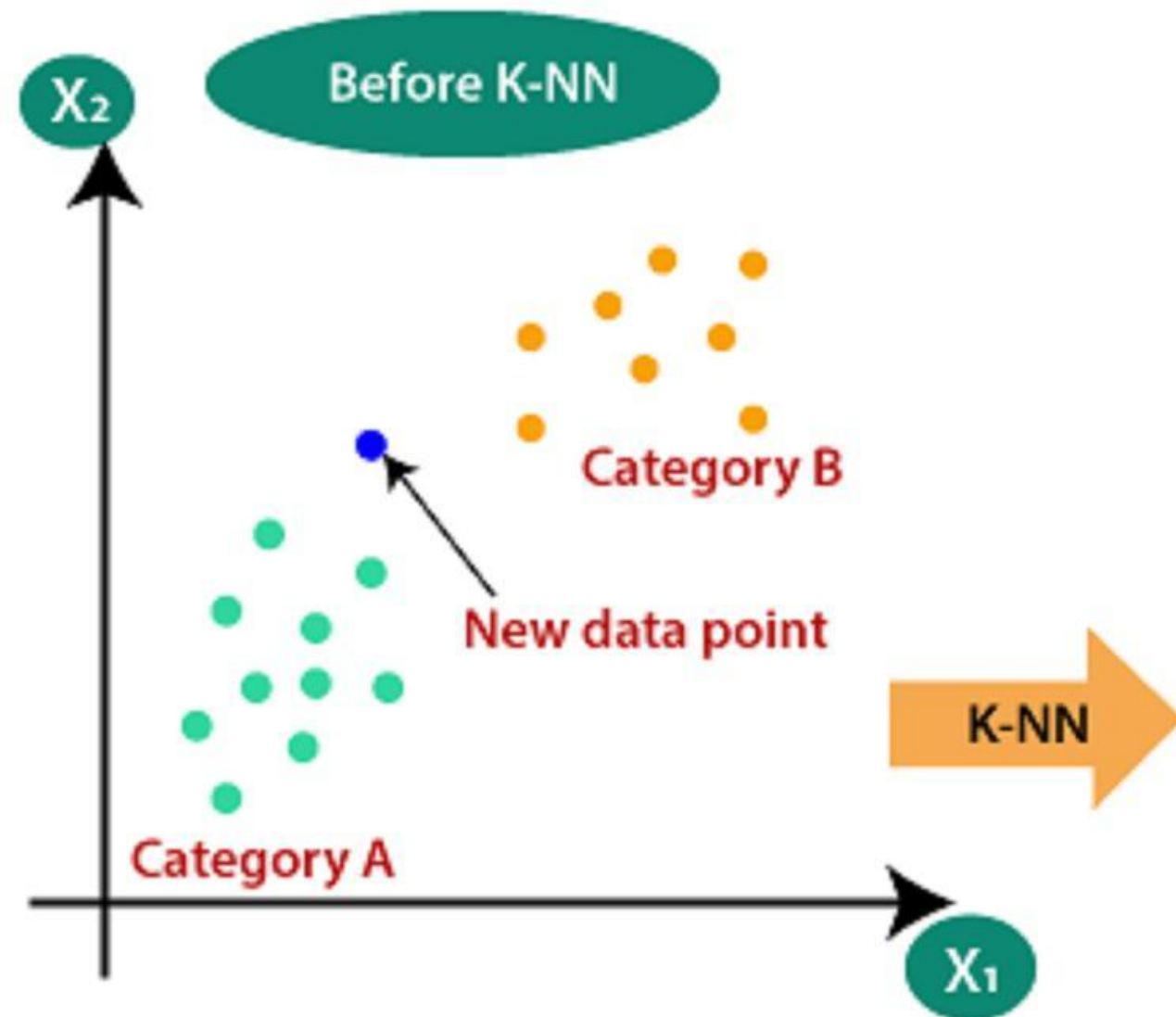
- If $K = 3$, the algorithm checks the 3 nearest points.

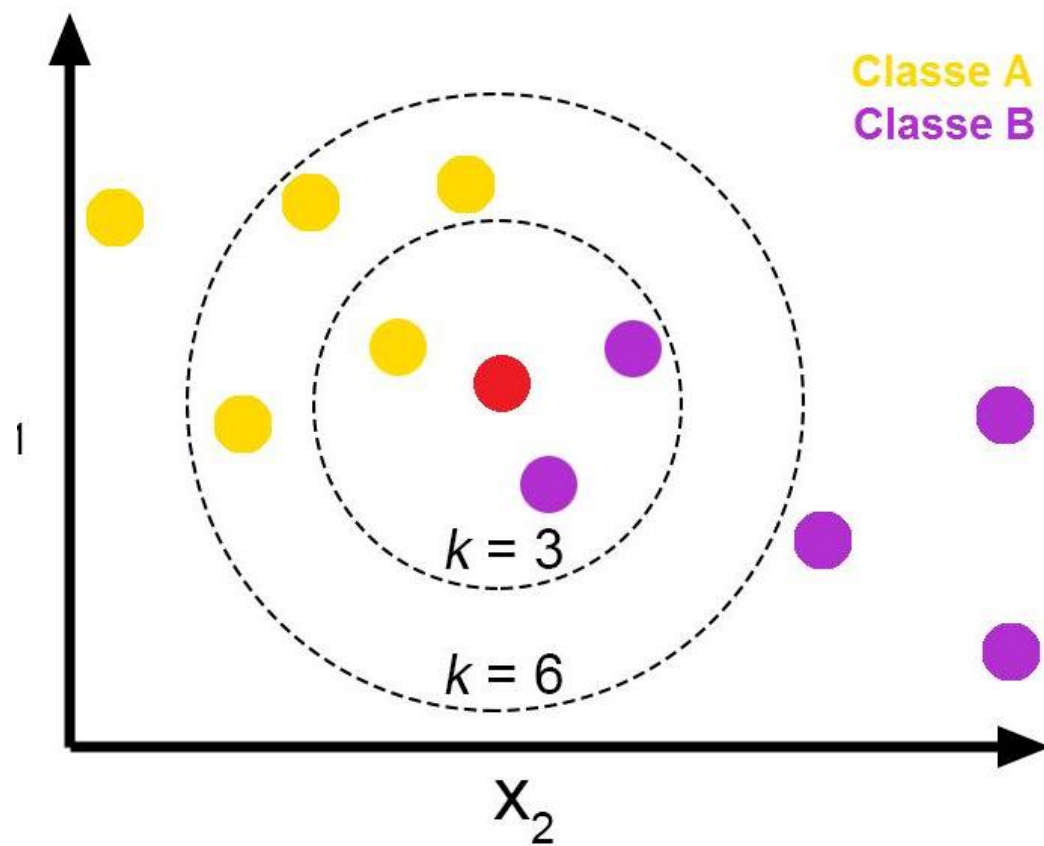
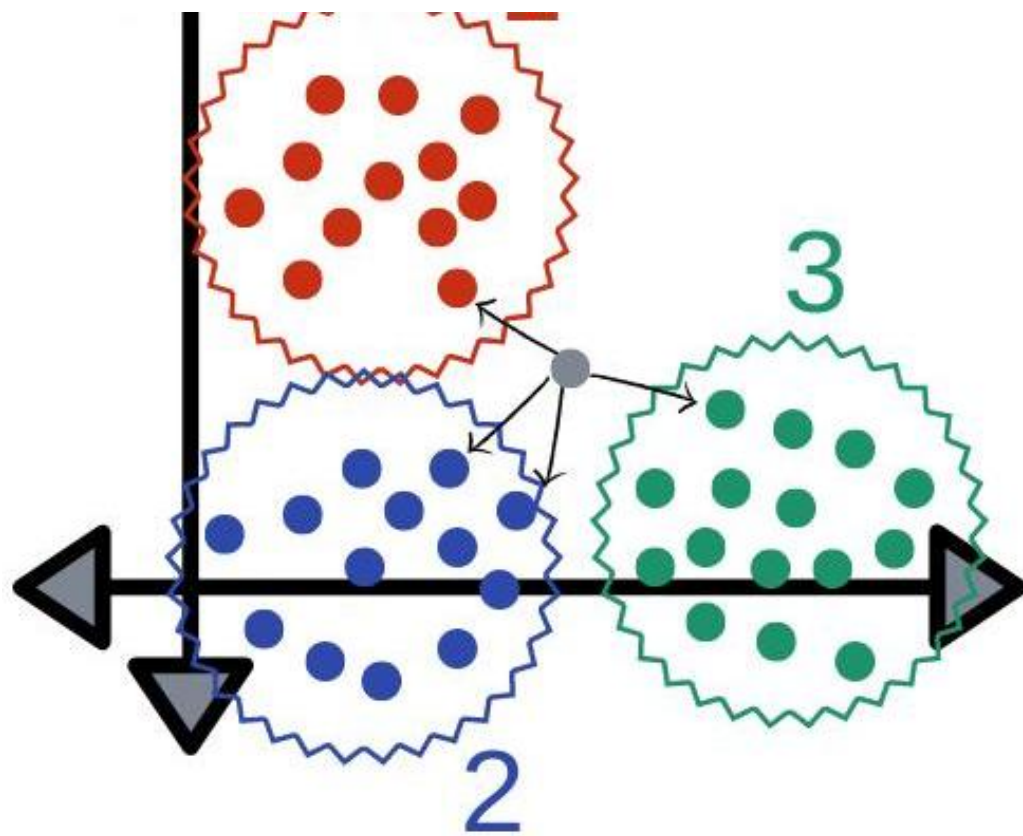
How KNN Works (Step-by-Step)

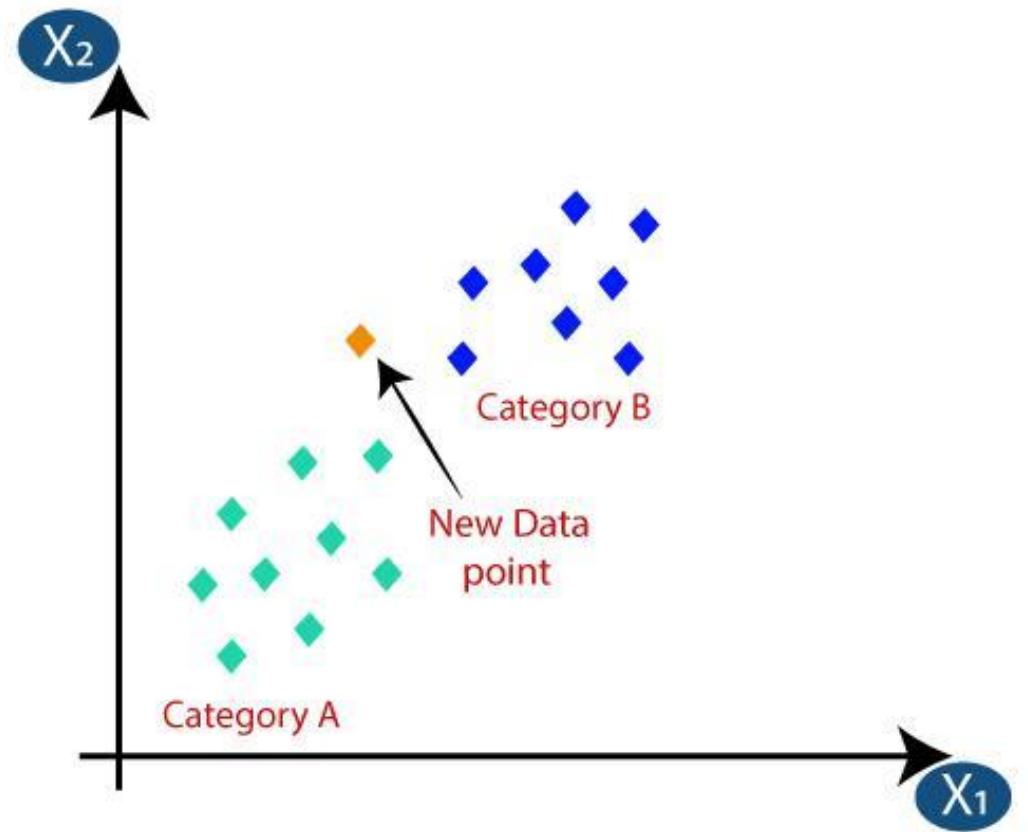
Given new point X:

1. Choose K
2. Calculate distance to all training data
3. Select K nearest points
4. Majority vote (classification)
5. Average value (regression)









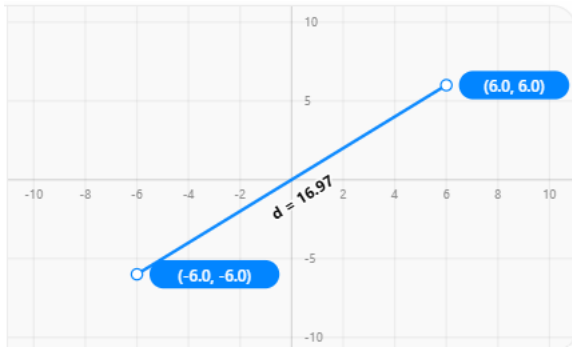
Distance Metrics in KNN

Distance determines similarity.

Common metrics:

- Euclidean Distance
- Manhattan Distance
- Minkowski Distance
- Hamming Distance

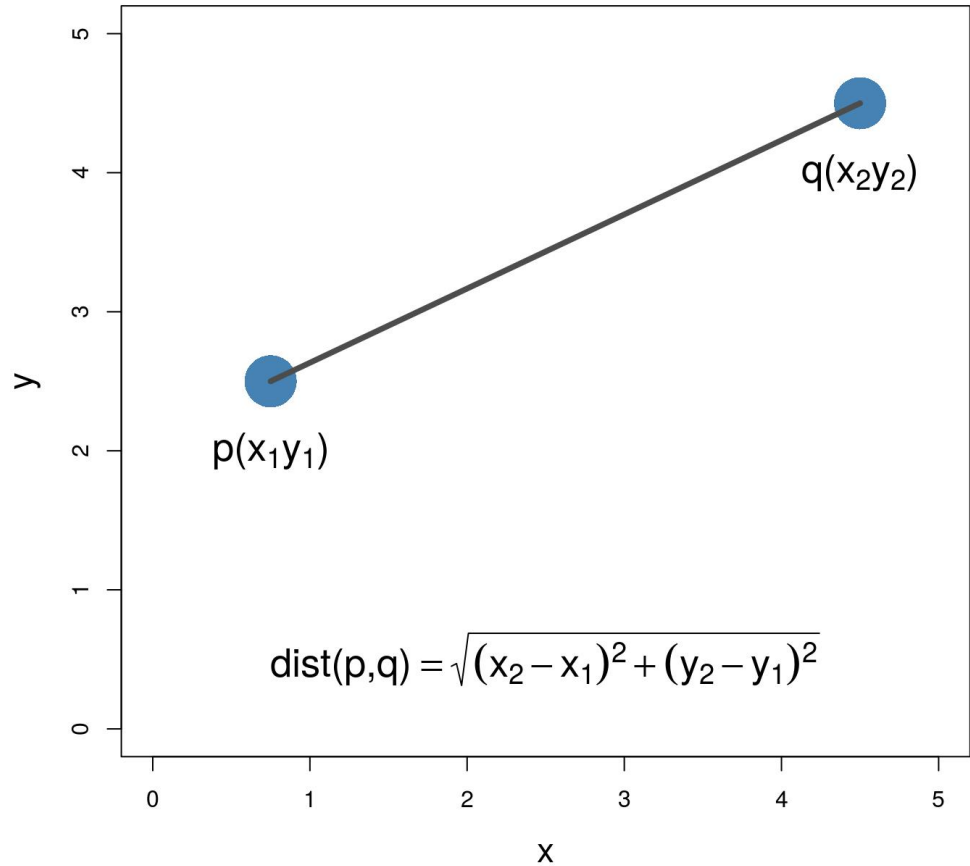
Euclidean Distance



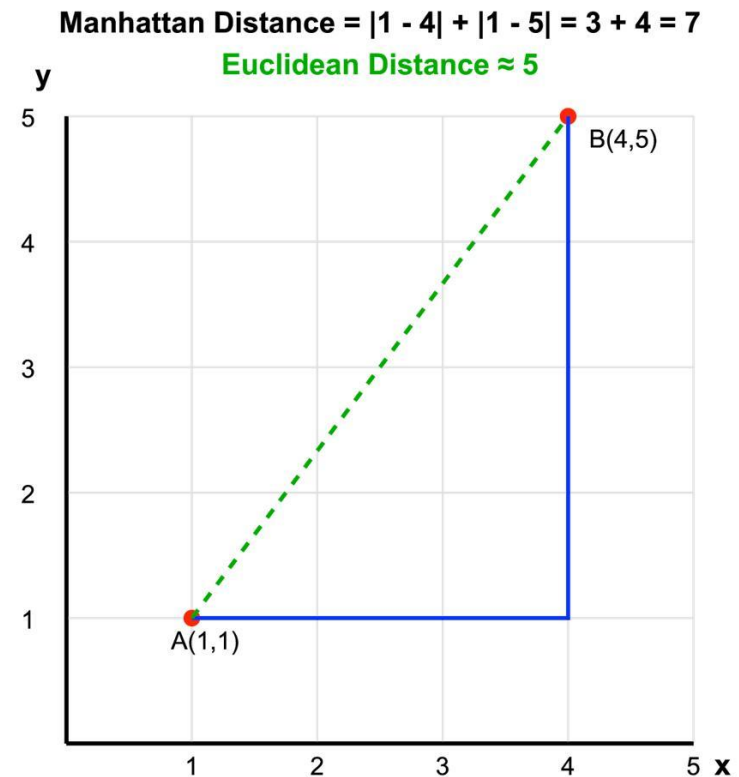
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- Straight-line distance between two points.

- Example:
- A(2,3), B(5,7)
- Distance:
- $= \sqrt{(5-2)^2 + (7-3)^2}$
- $= \sqrt{(9+16)}=5$



- **Manhattan Distance**
- Moves along grid lines.
- $d = |x_2 - x_1| + |y_2 - y_1|$
- Example:
- A(2,3), B(5,7)
- $= |5-2| + |7-3|$
- $= 3 + 4 = 7$



Choosing the Value of K

Small K:

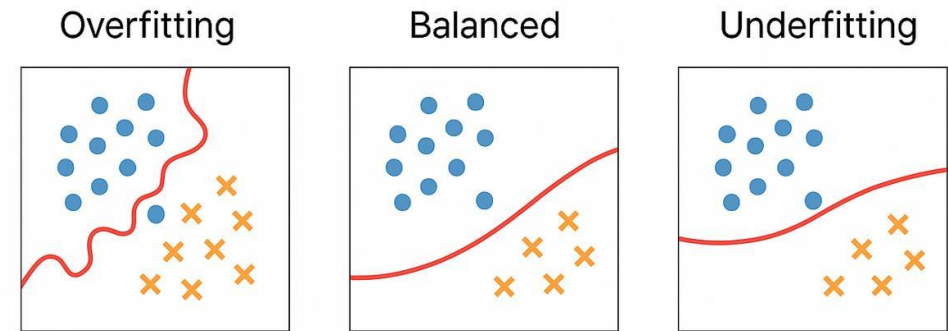
- Sensitive to noise
- Overfitting

Large K:

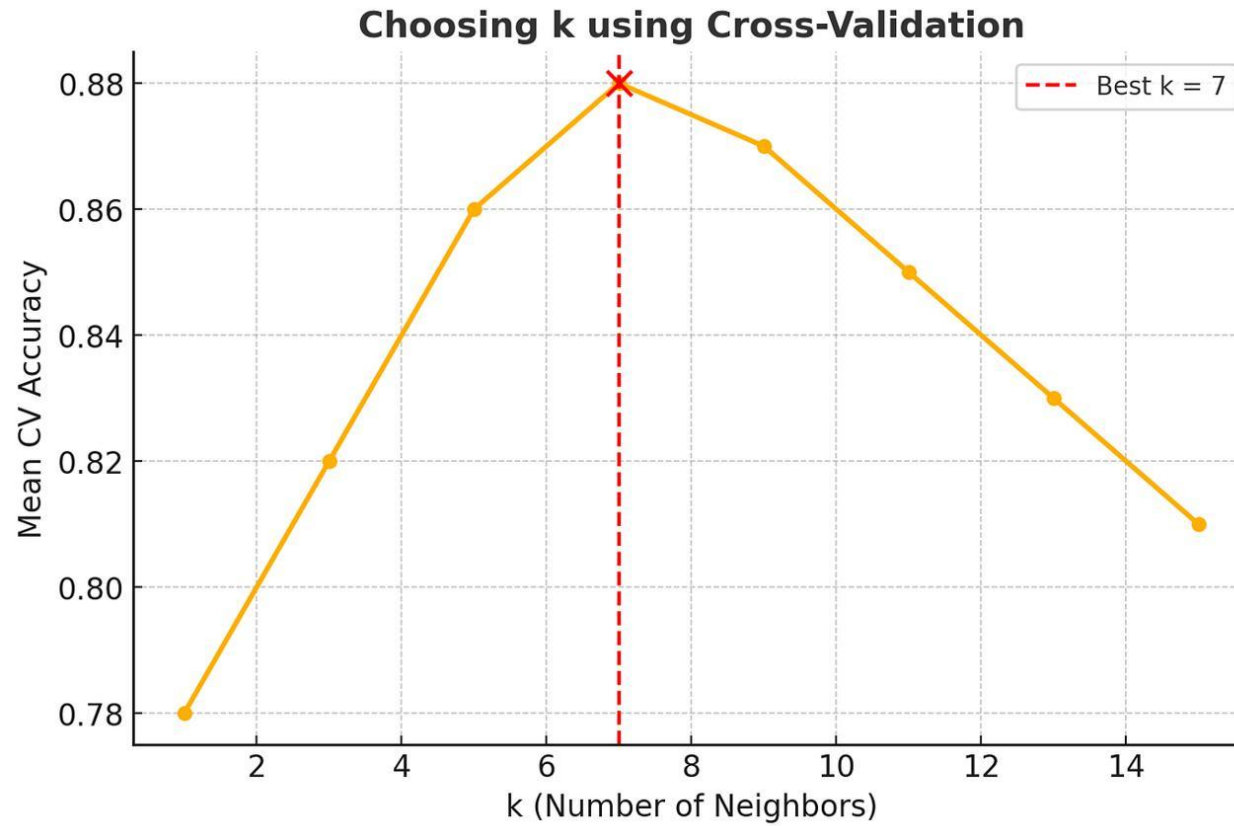
- Smooth decision boundary
- Underfitting

Best practice:

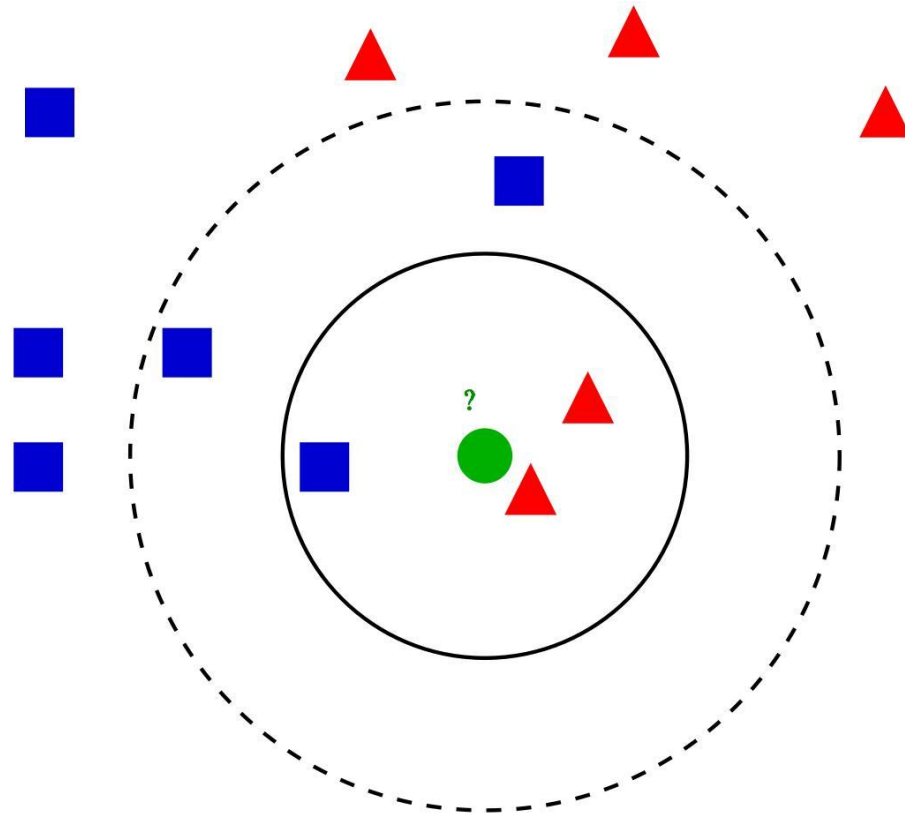
- Use odd K for binary classes
- Use cross-validation



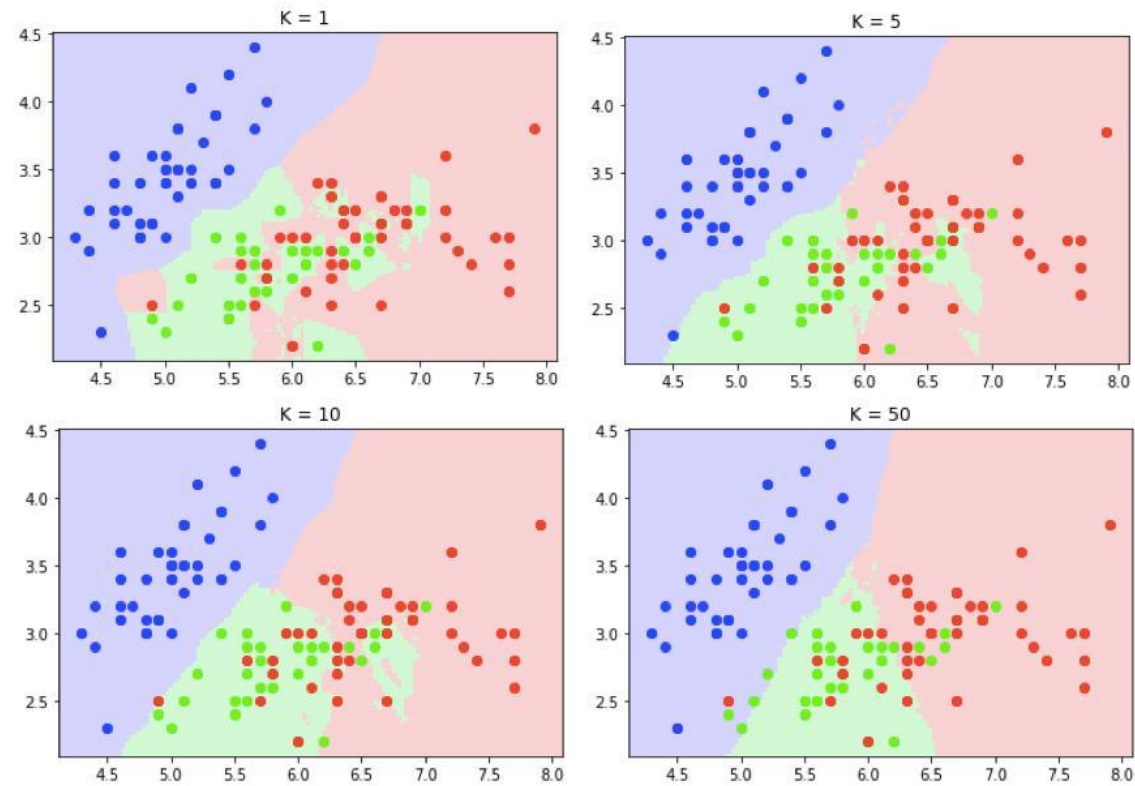
cross-validation



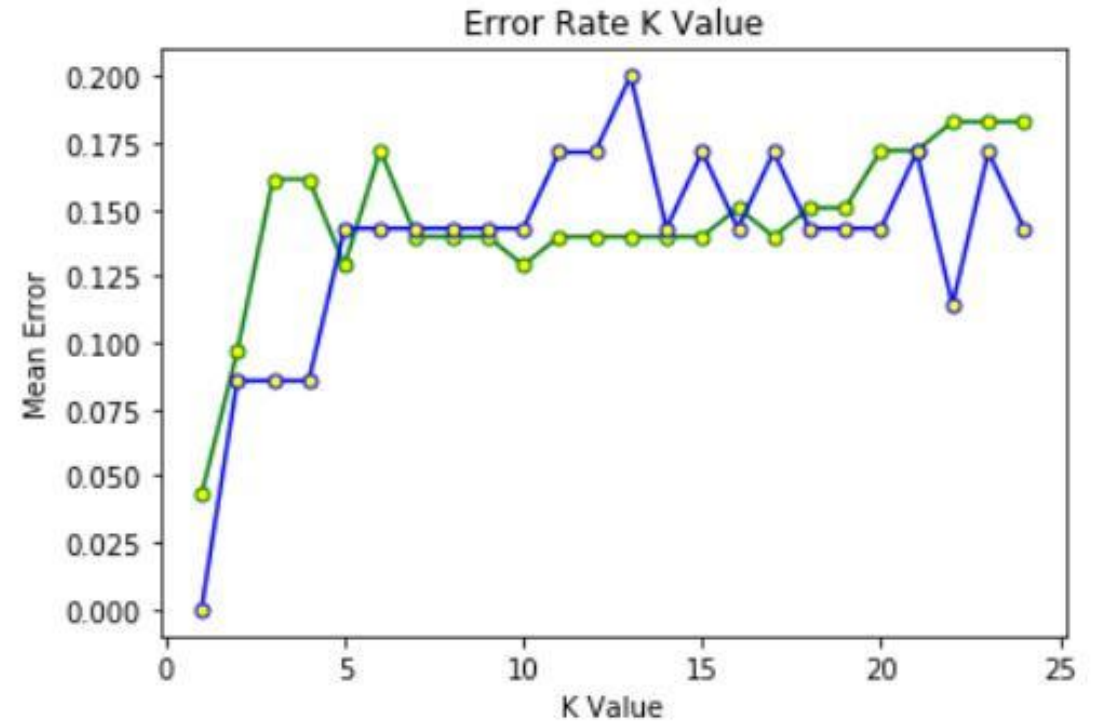
odd K for binary classes



odd K for binary classes



- The graph shows that $K=1$ gives lowest error but may overfit.
- Moderate K values such as 5–10 provide more reliable performance. Very large K increases bias and may reduce accuracy.



Bias vs Variance in KNN

K Value	Bias	Variance
Small K	Low	High
Large K	High	Low

Advantages of KNN

- Easy to understand
- No training phase
- Good for small datasets
- Works for classification & regression
- Flexible distance metrics

Limitations of KNN

- Slow for large datasets
- Sensitive to irrelevant features
- Needs feature scaling
- Memory expensive
- Poor with high dimensions